

SONAR-X

Physics-Aware Explainable Marine Debris Intelligence System

SURVEY INTELLIGENCE REPORT

Survey Information

Survey ID	SX-SRV-20260924-006
Survey Name	SONAR Survey — e2e sonar test
Survey Date	2026-09-24
Area	Not specified
Operator	Not specified
Data Source	SURVEY
Status	COMPLETED
Generated	2026-09-24 17:46 UTC
Software	SONAR-X 0.1.0-prototype

Sonar Metadata

Altitude above seabed	Not available
Range scale	Not available
Channel layout	BOTH
Already gain corrected	No
Already slant-range corrected	No

Detection Summary

Total detections	5
Marine Debris	0
Natural Formation	0
Sonar Artifact	0
Unknown Anomaly	5
Hazard — High Caution	4
Hazard — Caution	1

Detection List

Detection UID	Class	Conf.*	Evidence*	Risk	Hazard	Lat	Lon	Status
SX-DET-SX-SRV-20260924-006-005	UNKNOWN ANOMALY	47	47	MEDIUM	CAUTION	N/A	N/A	PENDING

Detection UID	Class	Conf.*	Evidence*	Risk	Hazard	Lat	Lon	Status
SX-DET-SX-SRV-20260924-006-004	UNKNOWN ANOMALY	69	69	MEDIUM	HIGH_CAUTION	N/A	N/A	PENDING
SX-DET-SX-SRV-20260924-006-003	UNKNOWN ANOMALY	83	83	MEDIUM	HIGH_CAUTION	N/A	N/A	PENDING
SX-DET-SX-SRV-20260924-006-002	UNKNOWN ANOMALY	64	64	MEDIUM	HIGH_CAUTION	N/A	N/A	PENDING
SX-DET-SX-SRV-20260924-006-001	UNKNOWN ANOMALY	66	66	MEDIUM	HIGH_CAUTION	N/A	N/A	PENDING

* Confidence and Evidence Score are prototype scores, NOT validated probabilities.

■ HAZARD ASSESSMENT

The following detections triggered precautionary hazard flags. This system does NOT identify specific hazardous objects. "Potential Hazard — Do Not Disturb" means sonar evidence warrants human review. Absence of a hazard flag does NOT mean an object is safe.

● SX-DET-SX-SRV-20260924-006-005 — CAUTION — UNKNOWN_ANOMALY

Indicators: Regular / structured geometry, Limited metadata / reduced signal quality

● SX-DET-SX-SRV-20260924-006-004 — HIGH_CAUTION — UNKNOWN_ANOMALY

Indicators: Regular / structured geometry, Strong acoustic backscatter, Unknown class with strong acoustic evidence, Limited metadata / reduced signal quality

● SX-DET-SX-SRV-20260924-006-003 — HIGH_CAUTION — UNKNOWN_ANOMALY

Indicators: Regular / structured geometry, Strong acoustic backscatter, Unknown class with strong acoustic evidence, Limited metadata / reduced signal quality

● SX-DET-SX-SRV-20260924-006-002 — HIGH_CAUTION — UNKNOWN_ANOMALY

Indicators: Regular / structured geometry, Strong acoustic backscatter, Unknown class with strong acoustic evidence, Limited metadata / reduced signal quality

● SX-DET-SX-SRV-20260924-006-001 — HIGH_CAUTION — UNKNOWN_ANOMALY

Indicators: Large spatial extent, Regular / structured geometry, Strong acoustic backscatter, Unknown class with strong acoustic evidence, Limited metadata / reduced signal quality

Limitations and Disclaimers

- This is a prototype system. All detections require human verification before any action.
- Confidence and evidence scores are prototype heuristic values, NOT validated probabilities.
- Side-scan sonar does not directly provide water temperature or depth.
- Conventional side-scan sonar is not a direct bathymetric sensor.
- Slant-range correction geometry is approximate where metadata was unavailable.
- Geolocation is approximate. GPS coordinates require field calibration.
- A missing detection does NOT prove debris was removed or moved.
- Absence of a hazard flag does NOT mean an object is safe.

AI-generated observations require verification by qualified human experts.